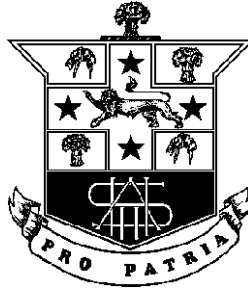


HURLSTONE AGRICULTURAL HIGH SCHOOL



MATHEMATICS – EXTENSION 2

2005

YEAR 12

HALF YEARLY EXAMINATION

Examiners ~ J Dillon, G Rawson

GENERAL INSTRUCTIONS

- | | |
|---|---|
| <ul style="list-style-type: none">• Reading Time – 5 minutes.• Working Time – 2 HOURS.• Attempt all questions.• All necessary working should be shown in every question.• This paper contains five (5) questions. | <ul style="list-style-type: none">• Marks may not be awarded for careless or badly arranged work.• Board approved calculators may be used.• Each question is to be started on a new piece of paper.• This examination paper must NOT be removed from the examination room. |
|---|---|

STUDENT NAME: _____

TEACHER: _____

Question 1 (15 marks) Start a NEW answer booklet

	Marks
(a) Let $u = 4 + 2i$ and $v = 3 - i$	
(i) Find $\bar{u}v$	1
(ii) Find v^2	1
(b) The complex number z satisfies both equations $ z - 1 = \frac{1}{2} z $ and $\arg(z - 1) - \arg z = \frac{\pi}{3}$	
(i) Show that $\frac{z - 2}{z} = \frac{1}{2} \operatorname{cis}\left(\frac{\pi}{3}\right)$.	2
(ii) Hence, show that $z = \frac{3 + i\sqrt{3}}{3}$.	2
(c) (i) In the same Argand diagram, sketch the locus of $ z - 3 = 1$ and the locus of $ z = z - 2i $.	2
(ii) Write down the complex number represented by the point of intersection of the two loci.	1
(d) (i) Find all solutions of the equation $z^3 = 1$. Leave your solutions in the form $z = \operatorname{cis} \theta$.	1
(ii) Let $\omega = \operatorname{cis} \frac{2\pi}{3}$. Show that $\omega^2 + \omega + 1 = 0$.	1
(iii) By expanding $(z + 2)(z^2 + z + 1)$, or otherwise, solve the equation $z^3 + 3z^2 + 3z + 2 = 0$.	2

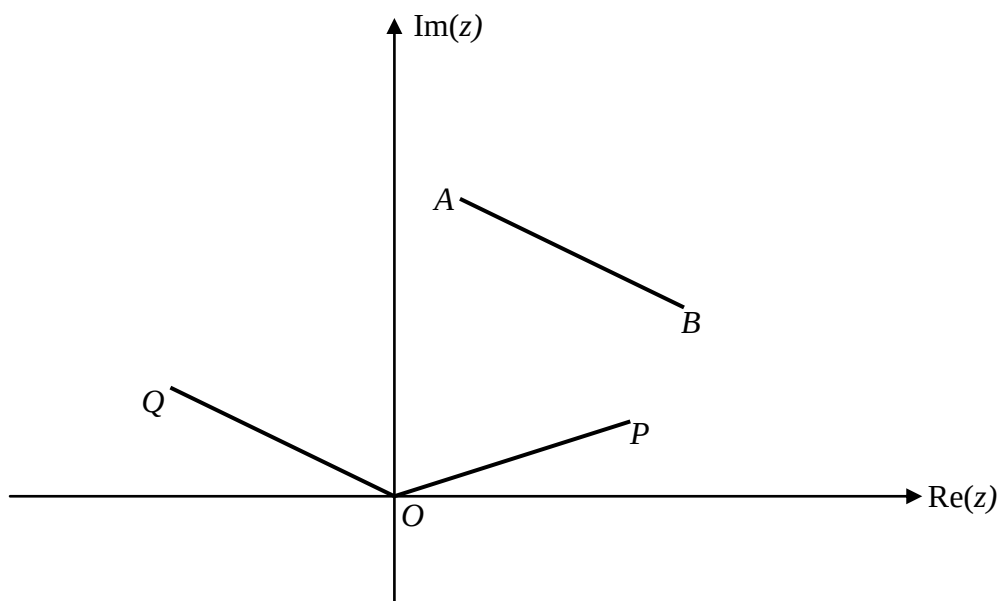
Question 1 continued on page 3

Question 1 (continued)

Marks

(e)

2



In the Argand diagram above, the intervals AB , OP and OQ are equal in length. Also, $QO \perp AB$ and $\angle POQ = \frac{2\pi}{3}$, where P has been rotated $\frac{2\pi}{3}$ anticlockwise about O to the point Q .

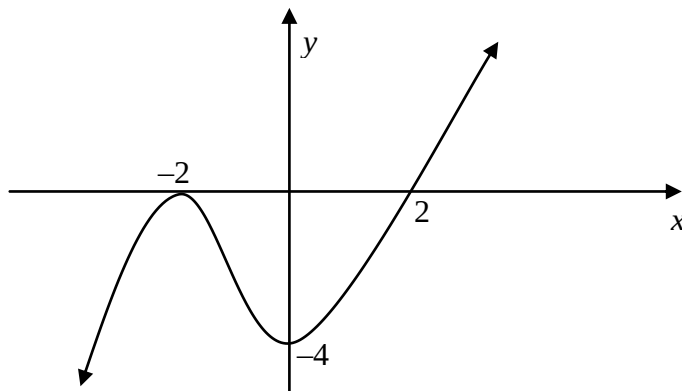
Let A and B represent the complex numbers u and v respectively.

Find in terms of u and v , the complex number represented by P .

Question 2 (15 marks) Start a NEW answer booklet

Marks

(a)



The curve shown in the diagram has equation $y = f(x)$.

There is a minimum turning point at $(0, -4)$ and a maximum turning point at $(-2, 0)$.

The curve crosses the x -axis at $(2, 0)$.

Sketch on separate diagrams:

(i) $y = |f(x)|$ 1

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y^2 = f(x)$ 2

(b) Let $f(x) = x - \frac{4}{x}$.

Provide separate half-page sketches of the graphs of the following functions.

(i) $y = f(x)$ 2

(ii) $y = \sqrt{f(x)}$ 2

(iii) $y = e^{f(x)}$ 2

Label each graph with its equation

(c) Find the equation of the tangent to the curve $xy(x + y) + 16 = 0$ at the point on the curve where the gradient is -1 .

4

Question 3 (15 marks) Start a NEW answer booklet

			Marks
(a)	(i)	Find all the zeros of $P(x) = x^4 - 5x^2 + 6$ over the rational numbers	2
	(ii)	Find $P(x)$, given that $P(x)$ is monic, of degree 4, with -1 as a single zero and 3 as a zero of multiplicity 3.	2
(b)		Find the remainder when $P(x) = x^3 + 2x^2 + 1$ is divided by $x + i$	2
(c)		The equation $x^3 + x^2 - 2x - 3 = 0$ has roots α , β and γ . Find the equations with roots	
	(i)	2α , 2β and 2γ	2
	(ii)	$\frac{\alpha}{2}$, $\frac{\beta}{2}$ and $\frac{\gamma}{2}$	2
	(iii)	$\alpha - 2$, $\beta - 2$ and $\gamma - 2$.	2
(d)		Consider $P(x) = x^4 + 3x^3 + 6x^2 + 12x + 8$,	
	(i)	Show that $x = 2i$ is a root of $P(x)$	1
	(ii)	Find all the roots of $P(x)$ in the complex field.	2

Question 4 (15 marks) Start a NEW answer booklet

Marks

Consider the ellipse $\frac{x^2}{16} + \frac{y^2}{12} = 1$.

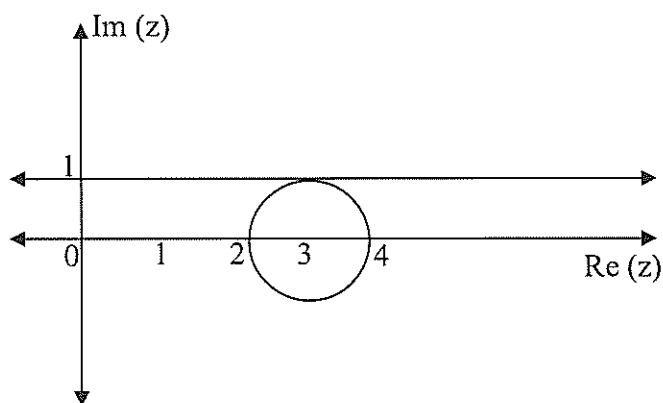
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|--------|--|----------|
| (i) | Find the eccentricity of the ellipse. | 1 |
| (ii) | Find the coordinates of the foci and the equations of the directrices of the ellipse. | 2 |
| (iii) | Sketch the graph of the ellipse showing clearly all of the above features and the intercepts on the coordinate axes. | 2 |
| (iv) | Use differentiation to derive the equations of the tangent and normal to the ellipse at the point (2, 3). | 3 |
| (v) | The tangent and normal to the ellipse at P cut the y – axis at A and B respectively. Find the coordinates of A and B . | 1 |
| (vi) | Show that AB subtends a right angle at the focus S of the ellipse. | 2 |
| (vii) | Show that the points A , P , S and B are concyclic. | 1 |
| (viii) | Find the centre and radius of the circle which passes through the points A , P , S and B . | 3 |

Question 5 (15 marks) Start a NEW answer booklet

Marks

- (a) Find real constants A and B such that $\frac{7x-4}{2x^2-3x-2} = \frac{A}{2x+1} + \frac{B}{x-2}$ **3**
- (b) Show that $(1 + \cos 2\theta + i \sin 2\theta)^n = 2^n \cos^n \theta (\cos n\theta + i \sin n\theta)$ **3**
- (c) Let $f(x) = \frac{x^2 - x}{x - 2}$
- (i) Show that $f(x) = x + 1 + \frac{2}{x - 2}$ **1**
- (ii) Using (i), or otherwise, determine the number of stationary points on $y = f(x)$. **2**
- (iii) Using (i), or otherwise, sketch the curve $y = f(x)$ indicating any asymptotes and x -intercepts. **3**
- (iv) Using your graph, or otherwise, solve the inequality $\frac{x^2 - x}{x - 2} > 0$. **1**
- (d) Sketch $y = |x| + |x + 2|$ for $-4 \leq x \leq 2$.
Hence solve the inequality $|x| + |x + 2| < 4$. **2**

Year12	Mathematics Extension 2	Half Yearly Examination 2005
Question No1	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E3 Uses the relationship between algebraic and geometric representations of complex numbers		
Outcome	Solutions	Marking Guidelines
E3 (a)(i)	$\begin{aligned}\bar{u}v &= (4-2i)(3-i) \\ &= 12-10i+2i^2 \\ &= 10-10i\end{aligned}$	Award 1 for correct answer
E3 (a)(ii)	$\begin{aligned}v^2 &= (3-i)^2 \\ &= 9-6i+i^2\end{aligned}$	Award 1 for correct answer
E3 (b)(i)	$\frac{ z-1 }{ z } = \frac{1}{2} \text{ so } \left \frac{z-1}{z} \right = \frac{1}{2}$ Also $\arg(z-1) - \arg(z) = \frac{\pi}{3}$ $\therefore \arg\left(\frac{z-1}{z}\right) = \frac{\pi}{3}$ $\therefore \frac{z-1}{z} = \frac{1}{2} \operatorname{cis} \frac{\pi}{3}$	Award 2 for correct solution Award 1 for reasonable attempt (failing to correctly show the result)
E3 (b)(ii)	$\begin{aligned}z-1 &= z\left(\frac{1}{2} \operatorname{cis} \frac{\pi}{3}\right) \\ z - z\left(\frac{1}{2} \operatorname{cis} \frac{\pi}{3}\right) &= 1 \\ z\left(1 - \frac{1}{2} \operatorname{cis} \frac{\pi}{3}\right) &= 1 \\ z\left(1 - \frac{1}{2} \cos \frac{\pi}{3} - \frac{1}{2} i \sin \frac{\pi}{3}\right) &= 1 \\ z\left(\frac{3}{4} - i \frac{\sqrt{3}}{4}\right) &= 1 \\ \therefore z &= \frac{1}{\frac{3}{4} - i \frac{\sqrt{3}}{4}} = \frac{1}{\frac{3}{4} - i \frac{\sqrt{3}}{4}} \times \frac{\frac{3}{4} + i \frac{\sqrt{3}}{4}}{\frac{3}{4} + i \frac{\sqrt{3}}{4}} = \frac{3+i\sqrt{3}}{3}\end{aligned}$	Award 2 for correct solution Award 1 for a substantially correct attempt
E3 (c)(i)	$ z-3 =1$ represents a circle centre (3, 0) and radius 1 $ z = z-2i $ represents the perpendicular bisector of the interval joining (0, 0) and (0, 2)	Award 2 for correct diagram with correct point of intersection Award 1 for correct diagram but no attempt to find point of intersection, or incorrect diagram (with one mistake) and an attempt to find the point of intersection



E3 (c)(ii) The point of intersection represents the complex number $z = 3 + i$

Award 1 for correct answer

E3 (d)(i) Let $z = cis\theta$ ($|z| = 1$)
 $(cis\theta)^3 = 1$
 $\therefore cis3\theta = 1$ (by de Moivre's)

Hence, $3\theta = 0, 2\pi, 4\pi, \dots \Rightarrow \theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}, \dots$

$\therefore$ The roots are $1, cis\frac{2\pi}{3}, cis\frac{4\pi}{3}$.

Award 1 for correct solution

E3 (d)(ii) Roots are $cis0 = 1, cis\frac{2\pi}{3} = \omega, cis\frac{4\pi}{3} = \omega^2$

From equation $z^3 - 1 = 0$

Sum of roots $= 1 + \omega + \omega^2 = 0$

Award 1 for correct solution

E3 (d)(iii) $(z+2)(z^2+z+1) = z^3+3z^2+3z+2$
 $\therefore z^3+3z^2+3z+2 = 0$ has the same roots as
 $(z+2)(z^2+z+1) = 0$
 $\therefore$ Roots are $z = -2, \omega, \omega^2$

Award 2 for correct solution
 Award 1 for a substantially correct attempt

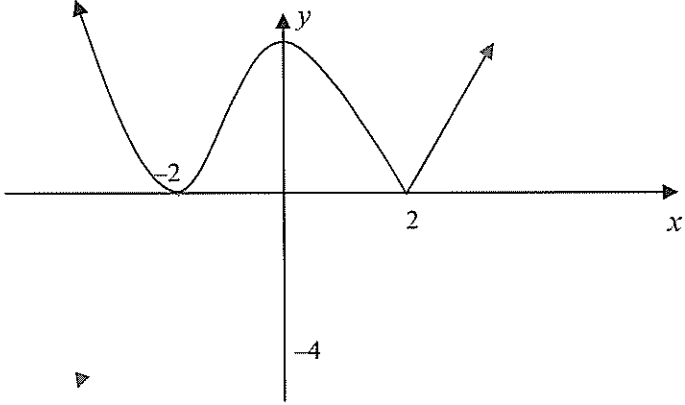
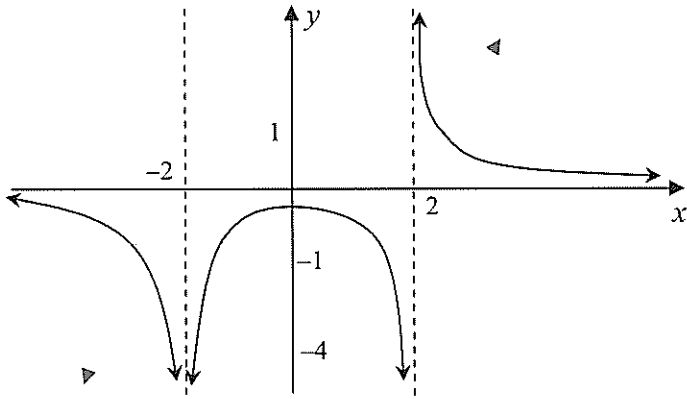
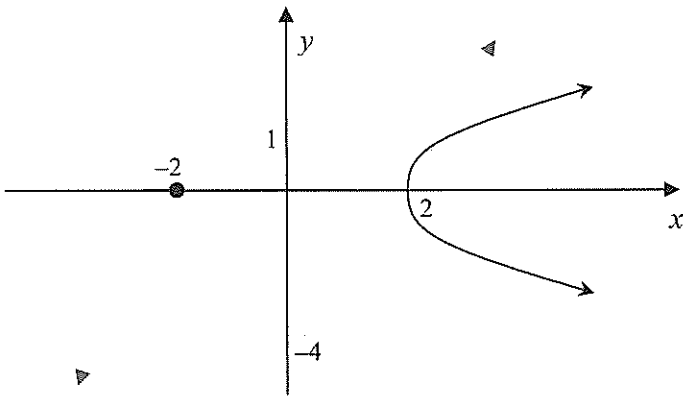
E3 (e) The vectors BA and OQ are equal. Hence, Q represents the complex number $z - u$.

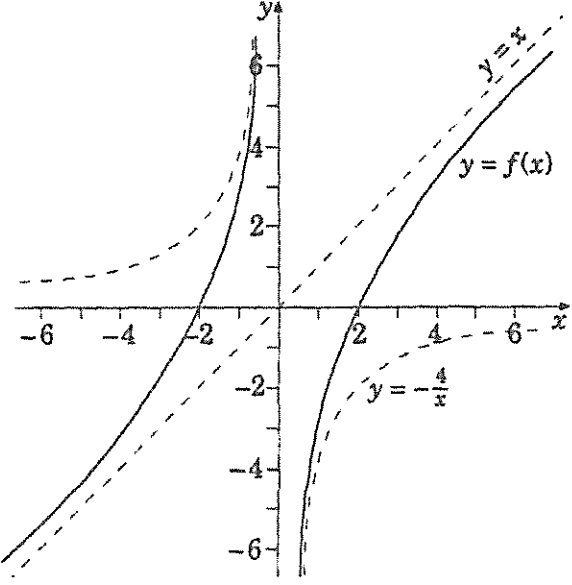
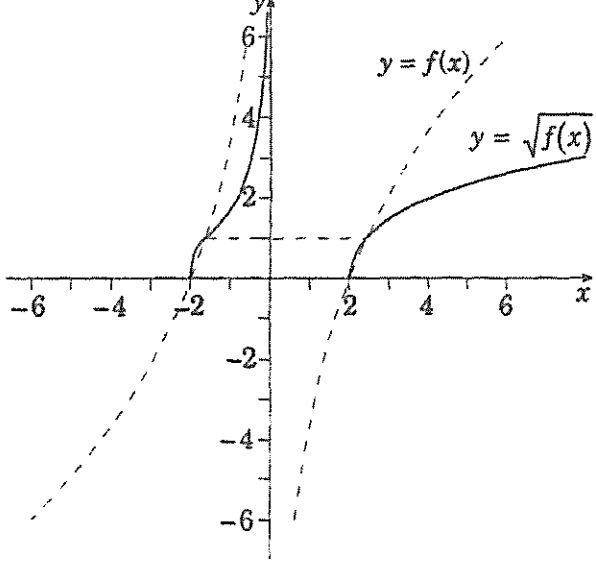
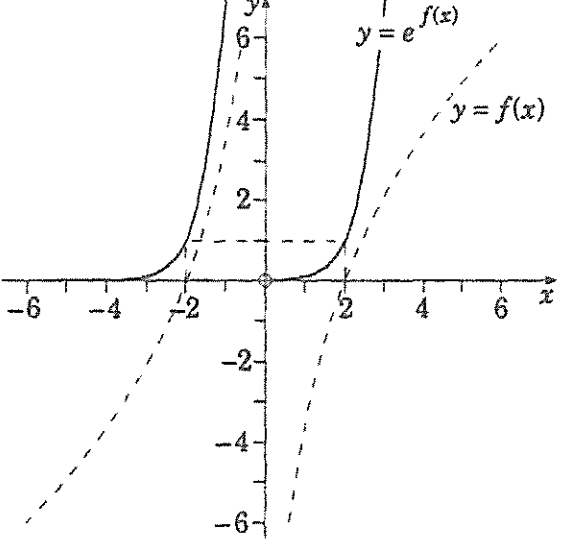
OP represents a clockwise rotation of OQ through $\frac{2\pi}{3}$.

$$\begin{aligned} \text{Hence } OP &= (u - v) \times cis\left(-\frac{2\pi}{3}\right) \\ &= (u - v) \left(\cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right) \right) \\ &= (u - v) \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2} \right) \end{aligned}$$

Award 2 for correct solution
 Award 1 for either $z - w$

$$\text{or } -\frac{1}{2} - i \frac{\sqrt{3}}{2}$$

Year 12	Mathematics Extension 2	Half Yearly Examination 2005
Question No. 2	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E6 combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions		
Outcome	Solutions	Marking Guidelines
E6 (a)(i)		<u>1 mark</u> : correct graph
E6 (a)(ii)		<u>2 marks</u> : correct graph with asymptotes OR <u>1 mark</u> : substantially correct attempt
E6 (a)(iii)		<u>2 marks</u> : correct graph with $(-2, 0)$ included OR <u>1 mark</u> : substantially correct attempt

Year 12	Mathematics Extension 2	Half Yearly Examination 2005
Question No. 2	Solutions and Marking Guidelines	
Outcome	Solutions	Marking Guidelines
E6 (b)(i)		<u>2 marks</u> : correct graph <i>with asymptotes</i> OR <u>1 mark</u> : substantially correct attempt
E6 (b)(ii)		<u>2 marks</u> : correct graph <i>with concavity changes</i> OR <u>1 mark</u> : substantially correct attempt
E6 (b)(iii)		<u>2 marks</u> : correct graph OR <u>1 mark</u> : substantially correct attempt

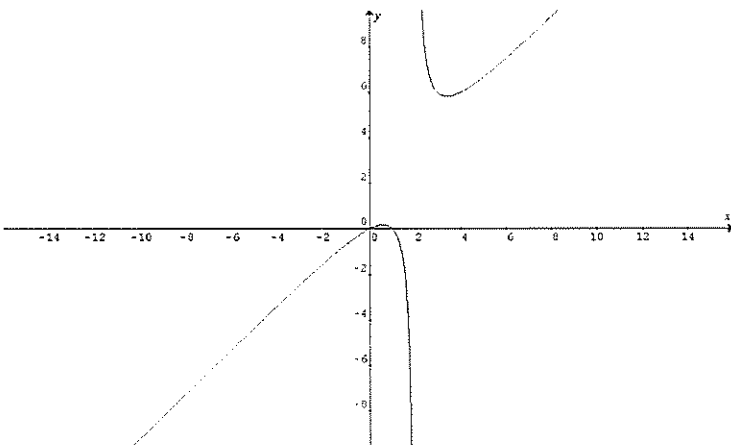
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Year 12	Mathematics Extension 2	HY Examination 2005
Question No. 3	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E4 uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials		
Outcome	Solutions	Marking Guidelines
E4 (a)(i)	$P(x) = x^4 - 5x^2 + 6$ $= (x^2 - 3)(x^2 - 2)$ $\therefore \text{no rational roots}$	<u>1 mark</u> : factorising $P(x)$ over rationals AND <u>1 mark</u> : stating no rational roots
E4 (a)(ii)	$Q(x) = (x+1)(x-3)^3$ $= x^4 - 8x^3 + 18x^2 - 27$	<u>2 marks</u> : $Q(x)$ in factorised form OR <u>1 mark</u> : having one factor correct with one factor degree 1 and the other factor degree 2.
E4 (b)	$P(x) = x^3 + 2x^2 + 1$ $P(-i) = (-i)^3 + 2(-i)^2 + 1$ $= i - 2 + 1$ $= i - 1$	<u>1 mark</u> : using the remainder theorem, with $x = -i$ AND <u>1 mark</u> : finding remainder $= i - 1$
E4 (c)(i)	$x^3 + x^2 - 2x + 1 = 0$ For roots $2\alpha, 2\beta, 2\gamma$, replace x with $\frac{x}{2}$ $\left(\frac{x}{2}\right)^3 + \left(\frac{x}{2}\right)^2 - 2\left(\frac{x}{2}\right) - 3 = 0$ $\frac{x^3}{8} + \frac{x^2}{4} - x - 3 = 0$ $x^3 + 2x^2 - 8x - 24 = 0$	<u>1 mark</u> : using correct substitution $\left(\frac{x}{2}\right)$ AND <u>1 mark</u> : obtaining correct equation (1 simple algebraic error allowed)
E4 (c)(ii)	For roots $\frac{\alpha}{2}, \frac{\beta}{2}, \frac{\gamma}{2}$, replace x with $2x$ $(2x)^3 + (2x)^2 - 2(2x) - 3 = 0$ $8x^3 + 4x^2 - 4x - 3 = 0$	<u>1 mark</u> : using correct substitution $2x$ AND <u>1 mark</u> : obtaining correct equation (1 simple algebraic error allowed)
E4 (c)(iii)	For roots $\alpha - 2, \beta - 2, \gamma - 2$, replace x with $x + 2$ $(x+2)^3 + (x+2)^2 - 2(x+2) - 3 = 0$ $x^3 + 7x^2 + 14x + 5 = 0$	<u>1 mark</u> : using correct substitution $(x+2)$ AND <u>1 mark</u> : obtaining correct equation (1 simple algebraic error allowed)

Year 12	Mathematics Extension 2	HY Examination 2005
Question No. 3	Solutions and Marking Guidelines	
Outcome	Solutions	Marking Guidelines
E4 (d)(i)	$P(x) = x^4 + 3x^3 + 6x^2 + 12x + 8$ $P(2i) = (2i)^4 + 3(2i)^3 + 6(2i)^2 + 12(2i) + 8$ $= 16 - 24i - 24 + 24i + 8$ $= 0 \quad \therefore \text{no remainder}$	<u>1 mark</u> : showing $P(2i)=0$
E4 (d)(ii)	coefficients of $P(x)$ are real $\therefore$ if $x - 2i$ is a factor, $x + 2i$ is also a factor <i>ie</i> $(x - 2i)(x + 2i) = x^2 + 4$ is a factor so $P(x) = x^4 + 3x^3 + 6x^2 + 12x + 8$ $= (x^2 + 4)(ax^2 + bx + c)$ $= (x^2 + 4)(x^2 + 3x + 2)$ $= (x - 2i)(x + 2i)(x + 2)(x + 1)$ $\therefore$ roots in the complex field are $-1, -2, 2i, -2i$	<u>1 mark</u> : factorising $P(x)$ at least over the real field <i>AND</i> <u>1 mark</u> : stating the actual complex roots

Year 12	Mathematics Extension 2	Half Yearly Examination 2005
Question No. 4	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E3	uses the relationship between algebraic and geometric representations of conic sections	
E4	uses efficient techniques for the algebraic manipulation required in dealing with questions involving conic sections	
Outcome	Solutions	Marking Guidelines
E3 (i)	$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{12}{16} = \frac{1}{4}$ $e = \frac{1}{2}$	Award 1 for correct answer
E3 (ii)	Foci = $(\pm ae, 0) = (\pm 2, 0)$ Directrices: $x = \pm \frac{a}{e} = \pm 8$	Award 1 for correct foci Award 1 for directrices
E3 (iii)		Award 2 for correct sketch showing all features and intercepts Award 1 for sketch showing 2 of the 3 requirements
E4 (iv)	$\frac{x^2}{16} + \frac{y^2}{12} = 1$ $\frac{2x}{16} + \frac{2y \frac{dy}{dx}}{12} = 0$ $\frac{dy}{dx} = -\frac{3x}{4y}$ At P(2, 3) $\frac{dy}{dx} = -\frac{1}{2}$ Equation of tangent is $y - 3 = -\frac{1}{2}(x - 2)$ $\Rightarrow x + 2y - 8 = 0$ Equation of normal is $y - 3 = 2(x - 2)$ $\Rightarrow 2x - y - 1 = 0$	Award 1 for determining $\frac{dy}{dx}$ Award 1 for equation of tangent Award 1 for equation of normal
E4 (v)	Tangent cuts y axis at A(0, 4) Normal cuts y axis at B(0, -1)	Award 1 for correct coordinates of both A and B
E4 (vi)	$m_{AS} = \frac{4-0}{0-2} = -2 \text{ and } m_{BS} = \frac{-1-0}{0-2} = \frac{1}{2}$ $\therefore m_{AS} \times m_{BS} = -1$ $\therefore \angle ASB = 90^\circ$	Award 2 for correct solution Award 1 for a solution that is substantially correct

E3 (vii)	$\angle APB = 90^\circ$ (tangent $\perp$ normal) $\therefore AB$ subtends equal angles of 90° at P and S $\therefore A, P, S$, and B are concyclic.	Award 1 for correct solution
E4 (viii)	Diameter of circle is AB Distance between A and $B = 5$ units Centre of circle is $\left(0, \frac{3}{2}\right)$ and radius is $\frac{5}{2}$ units	Award 3 for correct solution Award 2 for being able to identify AB as diameter and finding either centre or radius Award 1 for only finding centre, radius or diameter AB

Year12	Mathematics Extension 2	Half Yearly Examination 2005
Question No. 5	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
E3	Uses the relationship between algebraic and geometric representations of complex numbers	
E4	Uses efficient techniques for the algebraic manipulation required in dealing with questions involving polynomials	
E6	Combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions	
Outcome	Solutions	Marking Guidelines
E4 (a)	$\frac{7x-4}{2x^2-3x-2} = \frac{A}{2x+1} + \frac{B}{x-2}$ $\therefore 7x+4 = A(x-2) + B(2x+1)$ Let $x = 2$, $10 = 5B \Rightarrow B = 2$ Let $x = -\frac{1}{2}$, $-7\frac{1}{2} = -\frac{5}{2}A \Rightarrow A = 3$	Award 1 mark for correct value of A Award 1 mark for correct value of B
E3 (b)	$(1 + \cos 2\theta + i \sin 2\theta)^n = (1 + 2\cos^2 \theta - 1 + i.2\sin \theta \cos \theta)^n$ $= (2\cos \theta (\cos \theta + i \sin \theta))^n$ $= (2\cos \theta)^n (\cos \theta + i \sin \theta)^n$ $= 2^n \cos^n \theta (\cos n\theta + i \sin n\theta)$	Award 1 mark for use of trig. Identities Award 1 mark for factorisation Award 1 mark for use of de Moivre's Theorem
E6 (c)(i)	$f(x) = \frac{x^2-x}{x-2} = \frac{x^2-x-2+2}{x-2} = \frac{(x-2)(x+1)+2}{x-2}$ $= x+1 + \frac{2}{x-2}$	Award 1 mark for correct solution
E6 (c)(ii)	$f'(x) = 1 - \frac{2}{(x-2)^2}$ S.P's are where $f'(x) = 0$ $\therefore (x-2)^2 = 2$ $\therefore x = 2 \pm \sqrt{2}$ $\therefore$ There are two stationary points	Award 1 mark for correct derivative Award 1 mark for correctly identifying the number of SP's
E6 (c)(iii)		Award 3 marks for correct sketch showing both SP's, asymptotes and x intercepts Award 2 marks for sketch showing only 2 of the required features Award 1 mark for sketch showing only 1 feature

The x intercepts are $(0, 0)$ and $(1, 0)$

The asymptotes are

For $x \rightarrow \pm\infty$ $y = x + 1$

For $x \rightarrow 2^+$ $y = \frac{2}{x-2}$

For $x \rightarrow 2^-$ $y = \frac{2}{x-2}$

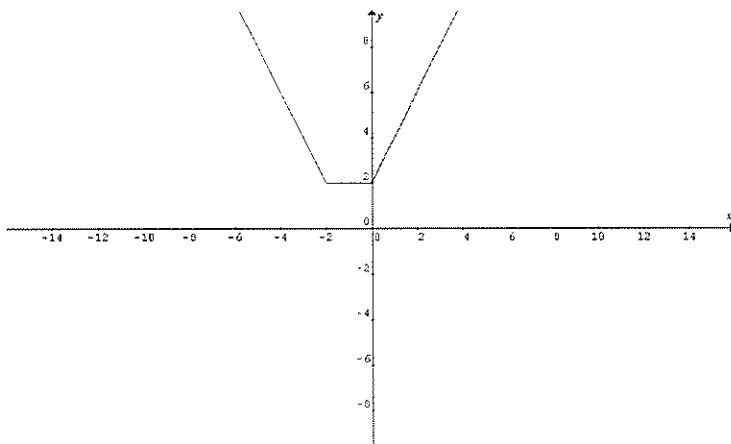
$x = 2$ is a vertical asymptote

E6 (c)(iv)

From the graph $\frac{x^2 - x}{x - 2} > 0$ when $0 < x < 1$ and $x > 2$

Award 1 mark for correct solution

E6 (d)



Award 2 marks for correct sketch

Award 1 mark for a sketch that is substantially correct

$|x| + |x + 2| > 4$ when $-3 < x < 1$

Award 1 mark for correct solution